Tagging Standards

> **Document Owner:** Platform Architecture Team | **Status:** Approved | **Last Updated:** February 2026

Overview

This document establishes enterprise-wide tagging standards for cost allocation, governance compliance, and operational visibility. Tags apply to:

- **Unity Catalog Securables** — Governed tags for catalogs, schemas, tables, and columns
- **Workflows** — Asset Bundle tags for jobs and pipelines
- **Serverless Compute** — Budget policies for notebooks, jobs, and model serving

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Golden Rules

ID	Rule	Severity	WAF Pillar
TG-01	All workflows (jobs, pipelines) must have required tags	Critical	Governance
TG-02	Use Governed Tags for UC securables (showback/chargeback)	Critical	Governance
TG-03	Serverless resources must use approved budget policies	Critical	Governance

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TG-01: Workflow Tags

All jobs and pipelines defined in Asset Bundles must include tags for cost allocation and operational tracking.

Required Tags

Tag	Required	Values	Description
`team`	Yes	Team name	Owning team for cost allocation
`cost_center`	Yes	CC-XXXX	Finance cost center code
`environment`	Yes	`dev`, `staging`, `prod`	Deployment environment
`project`	Recommended	Project name	Project or initiative name
`layer`	Recommended	`bronze`, `silver`, `gold`	Data layer
`job_type`	Recommended	`setup`, `pipeline`, `merge`, `ml`	Workload category

Job Tags Example

```
resources:
  jobs:
    gold_merge_job:
```

```
      name: "[${bundle.target}] Gold Merge Job"
      tags:
        team: data-engineering
        cost_center: CC-1234
        environment: ${bundle.target}
        project: customer-360
        layer: gold
        job_type: merge
```

Pipeline Tags Example

```
resources:
  pipelines:
    silver_pipeline:
      name: "[${bundle.target}] Silver Pipeline"
      tags:
        team: data-engineering
        cost_center: CC-1234
        environment: ${bundle.target}
        project: customer-360
        layer: silver
        pipeline_type: streaming
```

Querying Workflow Costs

```
SELECT
    usage_date,
    custom_tags:team AS team,
    custom_tags:cost_center AS cost_center,
    custom_tags:project AS project,
    SUM(list_cost) AS total_cost
FROM system.billing.usage
WHERE custom_tags:team IS NOT NULL
GROUP BY 1, 2, 3, 4
ORDER BY total_cost DESC;
```

TG-02: Unity Catalog Governed Tags

Governed Tags are account-level tags with enforced allowed values, ensuring consistent metadata across all workspaces.

Required Governed Tags

Tag	Apply To	Allowed Values	Purpose

Tag	Apply To	Allowed Values	Purpose
`cost_center`	Catalogs	Organization codes	Financial allocation
`business_unit`	Catalogs	Department names	Organizational grouping
`data_owner`	Schemas	Email or team name	Accountability
`data_classification`	Tables	`public`, `internal`, `confidential`, `restricted`	Security level
`pii`	Columns	`true`, `false`	PII indicator
`pii_type`	Columns	`email`, `phone`, `ssn`, `name`, `address`, `dob`, `financial`	PII categorization

Applying Tags by Level

Catalog-level:

```
ALTER CATALOG sales_data SET TAGS ('cost_center' = 'CC-1234');
ALTER CATALOG sales_data SET TAGS ('business_unit' = 'Sales');
```

Schema-level:

```
ALTER SCHEMA sales_data.gold SET TAGS ('data_owner' = 'analytics-
team@company.com');
```

Table-level:

```
ALTER TABLE gold.dim_customer SET TAGS ('data_classification' =
'confidential');
```

Column-level (PII):

```
ALTER TABLE gold.dim_customer
ALTER COLUMN email SET TAGS ('pii' = 'true', 'pii_type' = 'email');

ALTER TABLE gold.dim_customer
ALTER COLUMN phone SET TAGS ('pii' = 'true', 'pii_type' = 'phone');

ALTER TABLE gold.dim_customer
ALTER COLUMN ssn SET TAGS ('pii' = 'true', 'pii_type' = 'ssn');
```

Governed vs User-Defined Tags

Feature	Governed Tags	User-Defined Tags
Policy enforcement	Required	Optional
Predefined values	Enforced	Any value
Permission control	ASSIGN permission	Any user with MANAGE
Inheritance	To children	Manual only
Cross-workspace consistency	Account-level	Workspace-specific

Tag Inheritance

Governed tags inherit from parent to child objects:

```
Catalog (cost_center = CC-1234)
  └── Schema (inherits cost_center)
        └── Table (inherits cost_center)
              └── Column (inherits cost_center)
```

PII Type Reference

Type	Description	Column Examples
`email`	Email addresses	customer_email, contact_email
`phone`	Phone numbers	mobile_phone, work_phone
`ssn`	Social security numbers	ssn, tax_id
`name`	Full or partial names	customer_name, first_name
`address`	Physical addresses	street_address, mailing_address
`dob`	Date of birth	birth_date, dob
`financial`	Financial account numbers	account_number, credit_card

Querying Governed Tags

Find all PII columns:

```
SELECT
    table_catalog,
    table_schema,
    table_name,
    column_name,
    tag_value AS pii_type
```

```
FROM system.information_schema.column_tags
WHERE tag_name = 'pii_type';
```

Tables by classification:

```
SELECT
    table_catalog,
    table_schema,
    table_name,
    tag_value AS data_classification
FROM system.information_schema.table_tags
WHERE tag_name = 'data_classification';
```

Cost centers by catalog:

```
SELECT
    catalog_name,
    tag_value AS cost_center
FROM system.information_schema.catalog_tags
WHERE tag_name = 'cost_center';
```

TG-03: Serverless Budget Policies

Serverless compute (notebooks, jobs, pipelines, model serving) uses budget policies for cost allocation and governance.

Setting Up Budget Policies

1. Navigate to **Settings** → **Compute** → **Serverless budget policies**
2. Create policy with required tags (team, cost_center, environment)
3. Assign **User** or **Manager** permission to groups
4. Users select policy when creating serverless resources

Budget Policy Tags

Budget policies should enforce:

Tag	Required	Description
`team`	Yes	Team name for cost allocation
`cost_center`	Yes	Finance cost center code
`environment`	Yes	dev, staging, or prod
`project`	Recommended	Project or initiative

Querying Serverless Costs

```sql
SELECT
    usage_date,
    sku_name,
    custom_tags:team AS team,
    custom_tags:cost_center AS cost_center,
    SUM(list_cost) AS total_cost
FROM system.billing.usage
WHERE sku_name LIKE '%SERVERLESS%'
  AND custom_tags:cost_center IS NOT NULL
GROUP BY 1, 2, 3, 4
ORDER BY usage_date DESC, total_cost DESC;
```

Validation Checklist

Workflow Tags

- [ ] All jobs have `team` tag
- [ ] All jobs have `cost_center` tag
- [ ] All jobs have `environment` tag (using `${bundle.target}`)
- [ ] All pipelines have required tags

Governed Tags

- [ ] Catalogs tagged with `cost_center` and `business_unit`
- [ ] Schemas tagged with `data_owner`
- [ ] Confidential tables tagged with `data_classification`
- [ ] All PII columns tagged with `pii = true`
- [ ] All PII columns tagged with `pii_type`

Serverless Budget Policies

- [ ] Budget policies created for each team/cost center
- [ ] Required tags enforced in policy
- [ ] Teams assigned appropriate permissions
- [ ] Policy selection required for serverless resources

Quick Reference

Required Tags by Context

Context	Required Tags
Jobs (Asset Bundle)	`team`, `cost_center`, `environment`
Pipelines (Asset Bundle)	`team`, `cost_center`, `environment`

Context	Required Tags
Catalogs (Governed)	`cost_center`, `business_unit`
Schemas (Governed)	`data_owner`
Tables (Governed)	`data_classification` (if confidential/restricted)
Columns (Governed)	`pii`, `pii_type` (if contains PII)
Serverless (Budget Policy)	`team`, `cost_center`, `environment`

Tag Query Patterns

```sql
-- Cost by team (all compute)
SELECT custom_tags:team, SUM(list_cost)
FROM system.billing.usage
GROUP BY 1;

-- PII column inventory
SELECT * FROM system.information_schema.column_tags
WHERE tag_name = 'pii';

-- Tables by classification
SELECT * FROM system.information_schema.table_tags
WHERE tag_name = 'data_classification';

-- Serverless cost by budget policy
SELECT custom_tags:cost_center, SUM(list_cost)
FROM system.billing.usage
WHERE sku_name LIKE '%SERVERLESS%'
GROUP BY 1;
```

Related Documents

- Data Governance
- Naming & Comment Standards
- Data Quality Standards
- Unity Catalog Tables
- Cluster Policies

References

- Unity Catalog Governed Tags
- Usage Attribution Tags
- Serverless Budget Policies